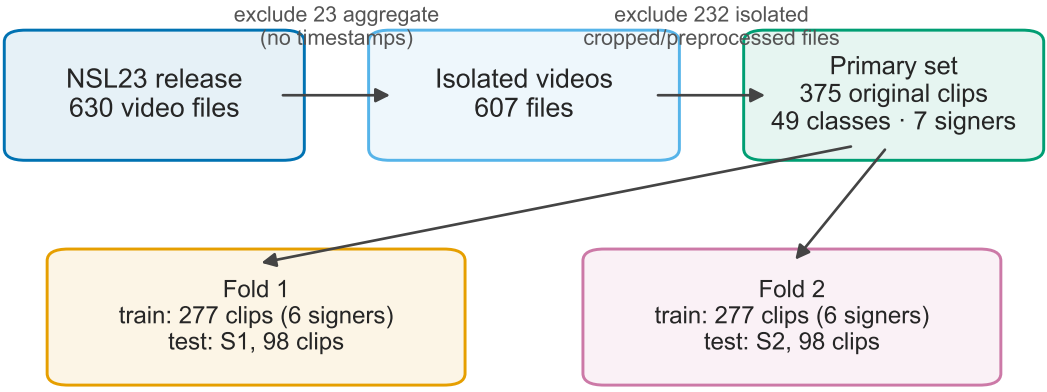


**5 models × 3 fixed seeds per fold = 30 runs**

Each held-out signer contributes all 49 classes in both bright and dark indoor conditions.

Aggregate and cropped/preprocessed categories overlap by two files; exclusions are shown sequentially.